

# Prepare Spatial Predictors for Model Fitting

## Description

Expands dataset-level spatial predictor data to the sample level by joining against the canonical '(dataset\_name, age)' sample index, producing a data frame whose rows align with the community matrix and abiotic data used for model fitting.

## Usage

```
prepare_spatial_predictors_for_fit(data_spatial = NULL, data_sample_ids = NULL)
```

## Arguments

**data\_spatial** A data frame with 'dataset\_name' as row names and one or more spatial predictor columns (e.g. 'coord\_x\_km', 'coord\_y\_km' from 'project\_coords\_to\_metric()').

**data\_sample\_ids** A data frame of valid '(dataset\_name, age)' pairs as returned by 'align\_sample\_ids()'.

## Details

Spatial predictors are stored at the dataset level (one row per site) but models require one row per sample (site  $\times$  time-slice). This function replicates each dataset's spatial values across all its valid sample ages. The row-name format "'<dataset\_name>\_\_<age>'" matches that of the community matrix and abiotic data frame. Unlike 'prepare\_coords\_for\_fit', this function is generic and imposes no assumptions on which spatial predictor columns are present.

## Value

A data frame with row names in the format "'<dataset\_name>\_\_<age>'" and the same predictor columns as 'data\_spatial'. Rows are sorted by 'dataset\_name' then 'age', matching the ordering of the community matrix and abiotic data produced by the respective preparation functions. Rows with any 'NA' in the spatial predictors are dropped.

## See Also

[project\_coords\_to\_metric()], [align\_sample\_ids()], [build\_jsdm\_fit\_input()], [prepare\_coords\_for\_fit()]